CHAGEE

# CHAGEE SINGAPORE

## FORECAST THE PEAK, SHARE THE COUNTER

PREDICTIVE AND PHYSICAL AI: WHERE TO OPEN NEXT, AND WHAT THE GM DECIDES FOR THE BUSIEST HOUR

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# Agenda


01 **Chagee in Singapore**
How it earns and sells

02 **From farm to cup**
Where the supply chain is stable, and where it is not

03 **The data trail**
What every cup already records

04 **Our scope**
Two areas, two kinds of AI, one decision-maker

05 **Predict demand**
App searches, DBSCAN and a new-store pin

06 **Share the counter**
The bucket brigade, side by side

07 **Split the counter**
Which steps stay human: the GM's call

08 **The GM's playbook**
Four decisions and a pilot

09 **The size of the benefit**
Orders kept, revenue, and what it rests on




# In Singapore, outlets run under one company, and every channel ends at one counter

*One company and one data system: order and app data sit together (informal source, to confirm).*

HOW A SINGAPORE CUP IS SOLD · STEP THROUGH

GROUP NET REVENUE, FY 2025 · GROUP-WIDE, NOT SINGAPORE (SECONDARY SOURCE)




Franchise **88%** Company-owned **12%**, growing fastest

SG Singapore outlets run under one company and one data system (informal source, to confirm with Chagee)

| | |
|---|---|
| **In store** | Walk-in (app order; tourists can order at the counter, to confirm) |
| **Chagee app** | order ahead, collect |
| **Delivery** | third-party delivery apps (e.g. Grab); no Chagee-run delivery |

**One counter**
every channel's cup is made here




44 outlets, ground-floor mall units by preference; kiosks in basements and MRT stations at entry in 2019

- **One company, one data system**
  Singapore outlets run under one company and one data system. Our reading: a counter design or roster rule could be rolled out across outlets.
  A Chagee contact (informal), to confirm with Chagee
- **App-first**
  Most orders come through the app, with few kiosks in store. Orders are recorded; whether searches and customer location are logged and usable is to confirm.
  Team observation, to confirm
- **Why Chagee**
  A fast-growing, playful brand with heavy app use: likely rich order data, and room for AI to help the stores.

**TAKEAWAY** | Whatever the channel, every order ends at one counter. The peak hour is where an outlet wins or loses.

# Supply is standard and planned ahead; the counter feels demand within minutes

*Tea leaves travel one standard route. Only fresh milk and fruit are bought locally.*

FOLLOW ONE CUP · STEP THROUGH



### Source

Tea leaves from Yunnan through standard channels for every market

DATA CREATED

Shipments, lead times

Local: fresh milk, fruit

### Warehouse

Separate central and regional warehouses keep storage and logistics costs down

DATA CREATED

Stock vs par, expiry

### Outlet counter

Machines brew each cup from a scanned recipe; people do ice, toppings and hand-over

DATA CREATED

Cups by hour

Wait times (to confirm)

### Last mile

Self-collect from the app, or delivery via third-party apps such as Grab; riders wait at the same counter

DATA CREATED

Orders by channel

Rider wait (to confirm)

### Customer

Orders in the app, collects at the counter, comes back (or not)

DATA CREATED

App orders, drop-offs

- **Stable upstream**
  Standard sourcing, regional warehouses and forecast-led restocking: Chagee already plans this part.
  YF Chuhai; Chagee Form 20-F
- **Volatile downstream**
  Demand at a counter moves by the hour: lunch, a mall event, rain, a new drink launch.
- **So we stop at the counter**
  The open question is not the shipment. It is how many people the counter needs this hour, and how they work.

**TAKEAWAY** | Chagee already plans its stock (public sources). How it rosters to the hourly peak is not public; that is the question we model.

# Orders are already recorded; the other inputs a forecast needs are candidates to confirm

*Four data sources and six decisions they could feed. We follow the four that start in the app and end at the counter.*

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

| | Store and production | App and members | Delivery | Supply and context |
|---|---|---|---|---|
| Demand forecasts | Cups per hour by drink and channel | Member repeat demand | Orders per hour by platform | Weather, holidays, launches |
| Staffing and prep | Wait times, machine logs | Pickup no-shows | Hand-over peaks | Labour hours vs cups per hour |
| Replenishment | Drink mix per outlet | Launch trial rates | Group-order toppings | Ingredient use, expiry, par |
| Queue and capacity | Wait time, footfall vs sales | Store-select drop-offs | Rider wait at counter | Stock-outs at peak |
| Site selection | Sales density per outlet | "No store nearby", home and work | Distance and fee by postcode | Mall footfall, competitors |
| Menu and promos | Drink share, customisation | Trial and repeat, voucher uplift | Delivery basket mix | Collab calendar |

White chips mark the core input; grey text refines it. Red rows: our scope. Wait times, footfall, searches ("no store nearby") and home and work location are candidates to confirm, not known fields.

- **Known and to confirm**
  Known: orders by channel and time (point of sale, app, Grab). To confirm: searches, customer location, wait times and walk-aways; some may need new logging.
- **The app is the link**
  Only the app ties a cup to a person. That needs consent and care under Singapore's data protection law (PDPA).
- **What we take forward**
  Demand forecast, then staffing, then queue capacity: one chain that ends with the GM's counter and roster.

**TAKEAWAY** | Orders by hour are recorded; searches, location, waits and walk-aways need confirming. How Chagee forecasts and rosters today is not public; we show one way.

# New stores and the in-store peak, for one decision-maker: the GM

*Two of our three ideation areas, enabled by two kinds of AI, each simulated with the trade-off the GM weighs.*

FROM THREE IDEAS TO TWO · STEP THROUGH

NOT COVERED YET

### Online retailing

Steer orders between app pickup and Grab by hour.

**To be added.** Needs platform fees and price tests we do not have yet.

OUR FOCUS · EXPANSION

### New stores

Map where the next outlet adds the most cups, from app searches.

Rent, fit-out and payback stay future work.

Topic 3 · forecasting

OUR FOCUS · OPERATIONS

### In-store operations

Forecast the peak, set the roster, share the counter.

Order data exists; the decision is daily.

Topic 3 · forecasting

Topic 5 · bucket brigades

AI · PREDICTIVE

Forecast demand: where a new outlet adds cups, and cups per hour for the roster.

AI · PHYSICAL

Automation and machines at the counter, from dispensers up to machines that make the drink.

OPERATIONS TOOLS, NOT AI

Bucket brigade and cross-training: how people share the hand steps.

### The GM

General manager, Chagee Singapore

| | |
|---|---|
| DECIDES | Where to open next, which counter steps go to machines, how many people work the peak, and how they share it |
| WHEN | Before a store opens, and when rosters are set |
| USES | The forecasts and the simulations |
| TODAY | To confirm with Chagee: how rosters are set now |

SIMULATED, WITH THE TRADE-OFF THE GM WEIGHS

| | |
|---|---|
| PREDICT | cups added vs cups moved from nearby outlets |
| SHARE | bucket brigade vs fixed posts |
| SPLIT · MACHINES | shorter wait vs higher cost |
| BENEFIT | orders kept vs extra labour |

**TAKEAWAY** | Predictive AI says where to open and how many people the peak needs; physical AI changes which steps machines do; the bucket brigade decides who does the rest.

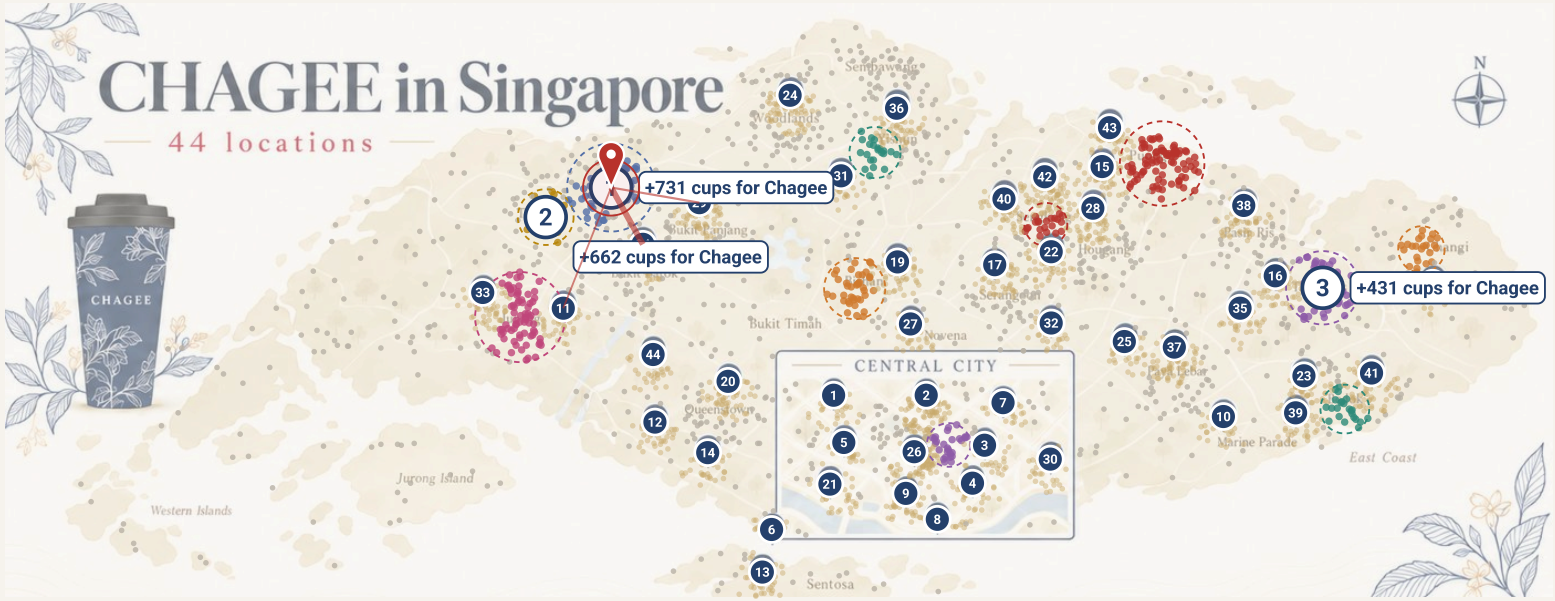
Course link · Week 3, Analytics and AI · the Flashion new-store forecast

# Where would outlet 45 add the most cups for Chagee?

*Not what a new store sells, but what it adds once you subtract the cups it takes from nearby outlets.*

**CONTEXT** Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential.

NEW-STORE FORECAST




**SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL**

**Strong site: mostly new cups**

**+731** cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

| | |
|---|---|
| 34 · West Mall, 2.7 km | **−77** |
| 29 · Bukit Panjang Plaza, 3.3 km | **−28** |

Peak hour 13:00: **92 cups** → **2 people** on today's counter (slides 06–07).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method.

**WITHOUT AI** our assumed baseline

Open outlet 45 beside 34 · West Mall: it sells **604** cups a day, but 64% are moved from outlets nearby, so **+216** for Chagee.

**WITH AI** our proposal

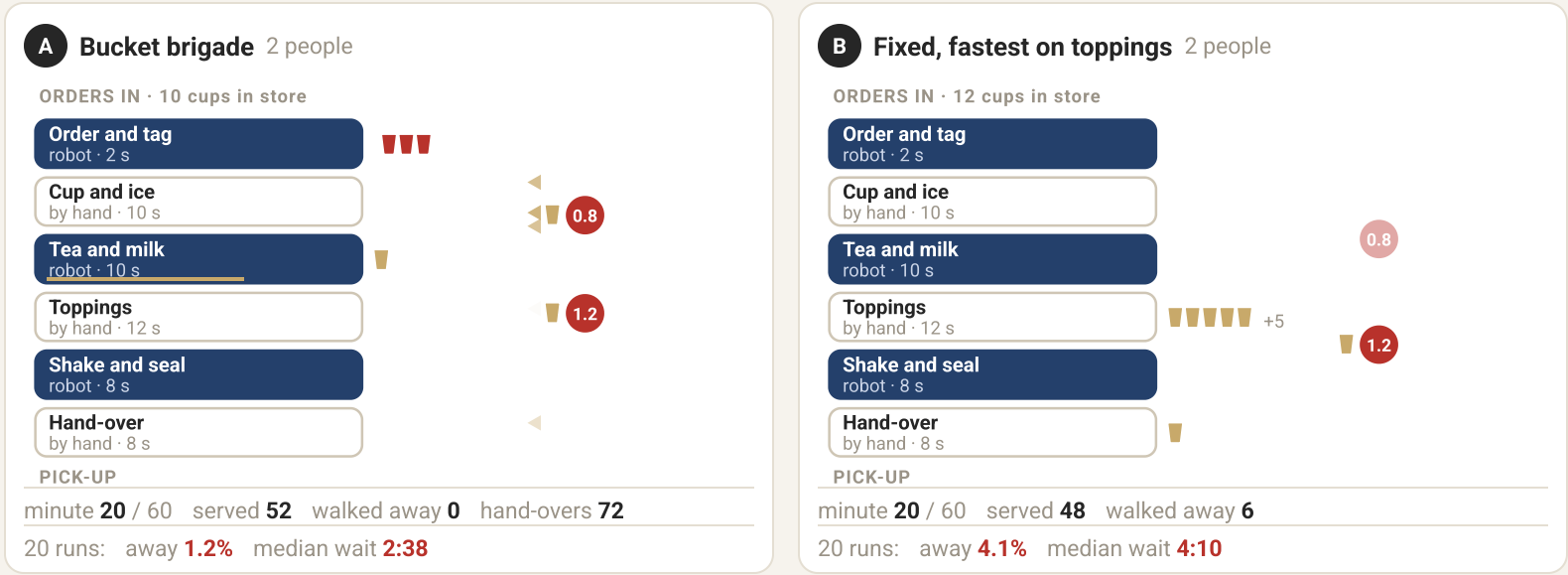
DBSCAN clusters synthetic app searches; the assumed demand model scores the same outlet 2.7 km away, near Choa Chu Kang: sells 837, **+731** for Chagee.

Course link · Week 5 and Week 6, equipment and technology · the bucket-brigade simulation

# With a lean roster, a bucket brigade keeps cups moving without knowing who is fastest

*Today's split (our scenario), two ways of sharing the hand steps, side by side. Pick what each side runs.*

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES




Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

ALL THREE RULES, MEAN OF 20 RUNS

| EVENT DAY, 2 PEOPLE | WALKED AWAY | MEDIAN WAIT |
|---|---|---|
| **Bucket brigade** | **1.2%** | **2:38** |
| Fixed, fastest on toppings | 4.1% | 4:10 |
| Fixed, slowest on toppings | 34.9% | 9:47 |

At the people on side A. A customer leaves if the wait would pass 5 min. With spare people, a well-assigned fixed line is as good or a little faster.

- **Why lean is where it counts**
  Work flows to whoever is free, so the line balances itself (Topic 5). Fixed stations are only as good as the guess of who is fast.

**ASSUMED BASELINE** *illustrative*
A normal-day roster (1 at the peak) on an event day: **54%** of peak-hour orders walk away.

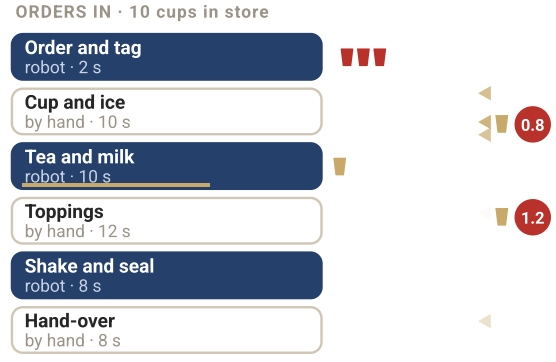
**PROPOSED SCENARIO** *in our model*
The forecast sets the crew (2) and a bucket brigade shares it: **1.2%** walk away, without knowing who is fastest.

Course link · Week 5 and Week 6, equipment and technology

# In our model, one tea-and-milk machine is the cheapest on every day type; pilot it against today's split

PEAK HOUR · SAME CUSTOMERS

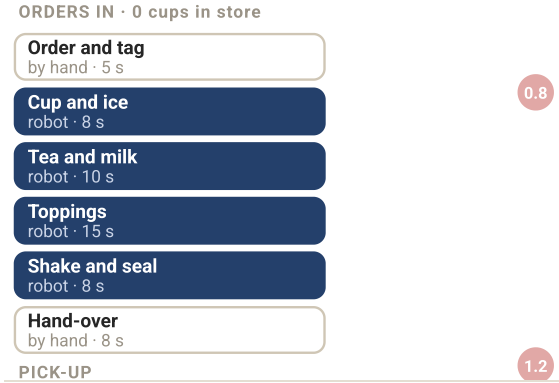
**A Today's split (our scenario)** 2 people, brigade




minute **20** / 60 served **52** walked away **0** hand-overs **72**

20 runs: away **1.2%** wait **2:38** **S$402**/day

**B Machine-made drink (scenario)** 2 people, fixed posts




minute **20** / 60 served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

FIVE OPTIONS, MEAN OF 20 RUNS

| EVENT DAY | PEOPLE | WALKED AWAY | WAIT | COST/DAY |
|---|---|---|---|---|
| Today's split (our scenario) | 2 | 1.2% | 2:38 | S$402 |
| Machine-made drink (scenario) | 2 fixed | 0.0% | 1:06 | S$465 |
| All by hand | 4 | 0.0% | 1:25 | S$422 |
| All robots | – | 0.0% | 0:58 | S$562 |
| **One machine (tea and milk) ✓** | **3** | **0.2%** | **1:44** | **S$335** |

✓ cheapest within the limit · fastest: All robots

**Normal day, cost a day:** One machine (tea and milk) S$305 (cheapest) · All by hand S$361 · Today's split (our scenario) S$372 · Machine-made drink (scenario) S$435 · All robots S$562.

ASSUMED BASELINE
illustrative

All by hand on an event day: **4 people**, median wait 1:24, **S$422** a day.

PROPOSED SCENARIO
in our model

In our model, one tea-and-milk machine on an event day: **3 people**, **S$335** a day, wait 1:44; today's split (our scenario) S$402, wait 2:38.

**GM DECISION** Pilot one tea-and-milk machine against today's split (our scenario): in our model S$305–335 a day against S$372–402. That holds only while a machine unit costs more than about S$905 a month (same rate for its 1 unit and the split's 3; wage, volume and crews held fixed). We assume S$1,909; Chagee's real cost is unknown.

# Four decisions for the GM, each from a simulation, then a pilot to test them

| | DECISION | WHY, FROM THE MODEL | REVISIT WHEN |
|---|---|---|---|
| 1 **Where to open** slide 05 · DBSCAN + assumed demand model | **Open where app searches cluster, not beside an existing outlet.** | **+731** against **+216** cups a day for the same new outlet. Synthetic app-search data: this shows the method, not a finding about real outlets. | Rerun on Chagee's app logs, if searches and location are recorded, before any lease. |
| 2 **What to automate** slide 07 · physical AI | **In our model, one tea-and-milk machine is the cheapest on every day type; pilot it against today's split (our scenario).** | **S$305–335** a day against S$372–402. The two cost the same at about **S$905 a month per machine unit** (1 unit against 3), with wage, volume, station speeds and crews held fixed; we assume S$1,909. Amacha's S$10,000 is a one-off price; without machine life, upkeep and financing we cannot turn it into a monthly cost. | When Chagee's real machine cost is known. |
| 3 **How many people** slides 06, 09 · forecast + economic rule | **Economic rule: add a person on event days, not on busy days.** | With the one machine: peak crew 2 → **3** on event days; 2 on busy days, where an extra person costs more than the orders it keeps. | When the forecast peak moves. |
| 4 **How to share** slide 06 · operations | **Test a bucket brigade against fixed posts for the hand steps.** | In the recommended setup (one machine, 3 people, event day): **0.2%** walk away with a brigade, 0.4% with well-assigned fixed posts, 22% with badly assigned ones; median wait 1:44 against 2:18. | Fixed posts once there is spare staff. |

- **Pilot in 3 outlets**

  **Shadow** forecast only | **Roster** economic rule | **Machine** one vs today's split

  KPIs: cups per labour hour, wait time and walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

  All figures are illustrative assumptions, not Chagee data. Machine costs and speeds are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

  How rosters are set today, the real machine cost per unit, and whether outlets log wait times and walk-aways.

**ASSUMED BASELINE** illustrative — One machine and a flat crew of 2 every day: **23%** of event-day peak orders walk away.

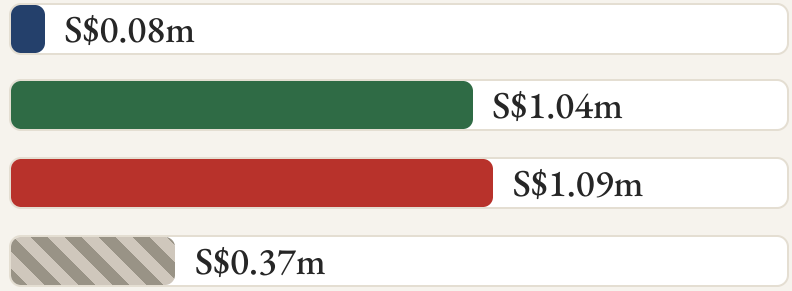
**PROPOSED SCENARIO** in our model — The forecast and the economic rule add a person on event days only (2 → 3): **0.2%** walk away; busy days stay at 2 (0.1%).

# In our model, about S$1.09m a year across 44 outlets against an illustrative baseline, almost all from one machine's labour saving

*Modelled avoided costs on stated assumptions, not demonstrated savings and not savings against Chagee's actual operations, which we do not know. The pilot compares one machine with today's split (our scenario).*

PROFIT A YEAR, ALL 44 OUTLETS · ECONOMIC RULE THROUGHOUT

AGAINST THE ILLUSTRATIVE ALL-HAND BASELINE (FLAT NORMAL-DAY ROSTER)




PILOT COMPARISON: THE PROPOSED SCENARIO AGAINST TODAY'S SPLIT (OUR SCENARIO), BOTH ON THE ECONOMIC RULE




Profit = orders kept × price × margin, plus labour saved, minus machine cost. Calendar: 104 busy days (weekends) and 11 event days (public holidays) a year, an illustrative definition. Baseline: all by hand, the normal-day crew every day. Peak hour only, so orders kept are a floor. Every figure, range and both crew rules: appendix.

PHYSICAL AI · ONE TEA-AND-MILK MACHINE

### Most of the benefit: labour

Lets the roster drop from 2 people to 1 in the off-peak hours: **S$1.04m** a year after the machine's cost.

FORECAST + ECONOMIC RULE

### Adds event-day orders on top of the machine

Adds a person on event days only (2 → 3), keeping about 35 more orders per outlet that day: 390 a year, **S$0.05m** across Singapore. On its own, all by hand: S$0.08m.

READ WITH CARE

### The big number rests on one assumption

The minimum crew, 2 by hand and 1 with machines, is ours. Without it, S$1.09m falls to **S$0.37m**. The pilot arm beats today's split by S$1.22m: more than the headline, because today's split is itself S$0.14m below the baseline.

**ASSUMED BASELINE** *illustrative*

Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

**PROPOSED SCENARIO** *in our model*

Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

# THANK YOU

## QUESTIONS AND DISCUSSION

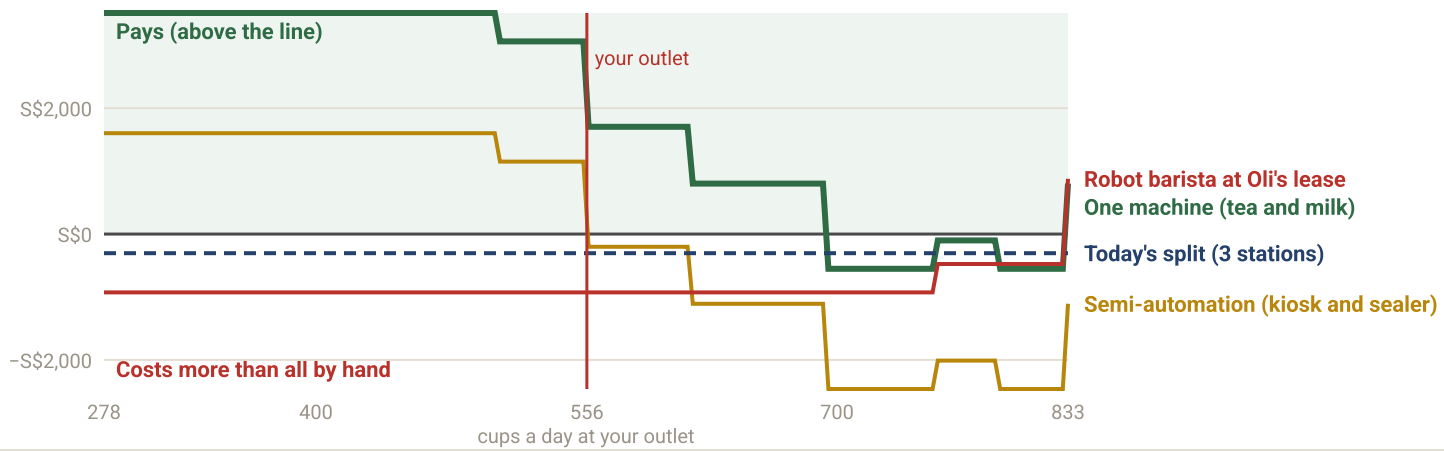
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# A machine pays only when it removes a person-shift for less than the shift costs

*To our knowledge, based on public sources; vendor list prices, not cost to implement. Chagee's equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).*

NET GAIN A MONTH AGAINST ALL BY HAND · ONE OUTLET




Shown at 556 cups a day, wage S$15.06 an hour, S$1,909 a month per machine unit (sliders on screen).

| | 1 Semi-automation | 2 Beverage platform | 3 Milk-tea machine | 4 Robot arm | 5 Robot barista |
|---|---|---|---|---|---|
| | Gong Cha Singapore | Gong cha Super Wu | Amacha, one shop | Richtech ADAM | Oli; coffee-first |
| | *Not disclosed* | *Not disclosed* | S$10,000 to buy | *Not disclosed, rental only* | US$4,990/mo lease |
| | PRESS | COMPANY CLAIM | PRESS | VENDOR PAGE | VENDOR PAGE |

The model's S$1,909 a month is per station unit (our reading: a semi-automated station). Semi-automation and the beverage platform are priced at it because their real prices are not disclosed; robot prices are per setup. Full table and sources in the appendix.

START HERE

**One machine on the tea and milk**

+S$1,706 a month vs all by hand

Saves 8 person-hours a day (S$3,614 a month) for one S$1,909 unit. Simple payback, if it lets the roster drop one person-shift: 2.8 months on Amacha's S$10,000.

WATCH

**Today's 3-station split**

−S$305 a month vs all by hand

Saves 12 person-hours a day (S$5,422) but pays for 3 units (S$5,726).

WAIT FOR VOLUME

**Robots: arm or barista**

757+ cups a day before they save more labour than today's split (labour only, not a cost break-even)

At 556 cups, Oli's lease (S$6,350) is S$928 a month worse; a robot setup would need to cost under S$5,422 a month. The arm's price is not disclosed.

**Model assumption that drives all three:** minimum crew 2 by hand, 1 with machines, off-peak. Station speeds are the model's, not vendor claims. Rosters step in whole people, so results flip with volume and wage.

WITHOUT AI
our assumed baseline

All by hand at 556 cups a day: **24 person-hours** a day, **S$10,843** a month of labour.

WITH AI
our proposal

One tea-and-milk machine: **16 person-hours** a day, **+S$1,706** a month after its unit cost. A robot setup (four stations on machines) starts to save more labour than today's split at about 757 cups a day; labour only, not a cost break-even.

# Per outlet, about 61 orders kept on an event day and 5 on a busy day; on an illustrative calendar, S$7,539 of revenue a year

*Modelled avoided costs on stated assumptions, not demonstrated savings, against an illustrative baseline, not Chagee's operations. Economic rule for the headline figures. Peak hour only; conservative. Prices to our knowledge, based on public sources.*

Profit a year, 44 outlets: **S$1.09m** = counter automation, physical AI **S$1.04m** (depends on the minimum-crew rule) + forecast roster, predictive AI **S$0.05m** | Without / with the minimum-crew rule: **S$0.37m to S$1.09m**

## ONE OUTLET AT 556 CUPS A DAY · BOTH LEVERS, ONE RUN

**EVENT DAY**
**61** orders kept
Revenue S$358 (× S$5.90) · profit S$241
Counter: labour saved − machine S$27

**BUSY DAY**
**5.4** orders kept
Revenue S$32 (× S$5.90) · profit S$76
Counter: labour saved − machine S$57

**NORMAL DAY**
**0.2** orders kept
Revenue S$1 (× S$5.90) · profit S$58
Counter: labour saved − machine S$57

| A YEAR | PER OUTLET | 44 OUTLETS, IF ALL LOOKED LIKE THIS MODEL |
|---|---|---|
| **Orders kept** peak hour, busy and event days | **1,278** range −121 to 1,452 | **56,221** |
| **Revenue** orders × price | **S$7,539** range −S$832 to S$10,015 | **S$331,704** |
| Profit margin 50%–70%: −S$754 to +S$754 | S$24,660 | S$1,085,027 |
| Labour saving, counter labour saved − machine cost; shown on its own | S$20,136 | S$886,005 |

Revenue and labour are never added: revenue is orders × price; profit is orders × price × margin, plus labour saved, minus machine cost.

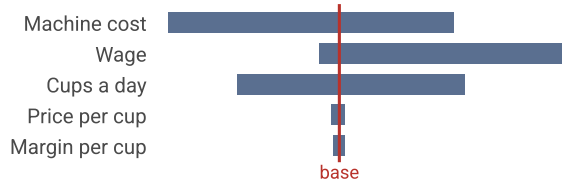
Calendar: 104 busy days (weekend days) and 11 event days (Singapore public holidays) a year: an illustrative definition, to confirm with Chagee; sliders on screen.

| ORDERS · PROFIT A YEAR | WALK-AWAY RULE | ECONOMIC RULE |
|---|---|---|
| Forecast crew | 1,242 · S$603 | 670 · S$1,711 |
| Machine | 888 · S$23,611 | 888 · S$23,611 |
| **Both, one run** | **1,278 · S$24,660** | **1,278 · S$24,660** |
| Both, today's split counter −S$305/month | 1,124 · −S$3,142 | 1,124 · −S$3,142 |

Walk-away rule: fewest people keeping peak-hour walk-aways within 2% (the forecast crew at 556 cups a day). Economic rule: add a person only when orders kept × price × margin beat the extra labour. "Both" is one run, not a sum: the levers keep many of the same orders, and with the machine the forecast crew needs fewer extra hours (−S$662 profit against the sum). Headline figures use the economic rule.

**Against what**
A flat normal-day roster and an all-by-hand counter. We do not know how Chagee sets rosters today (to confirm with Chagee).

**All inputs at their worst/best end together**
Profit a year, one outlet: low **−S$15,660** · base **S$24,660** · high **S$101,166**




**Fragile: the minimum crew**
The counter lever rests on the model's minimum crew off-peak: 2 by hand, 1 with machines. Profit a year per outlet: S$24,660 with it; S$8,395 with 1 either way.

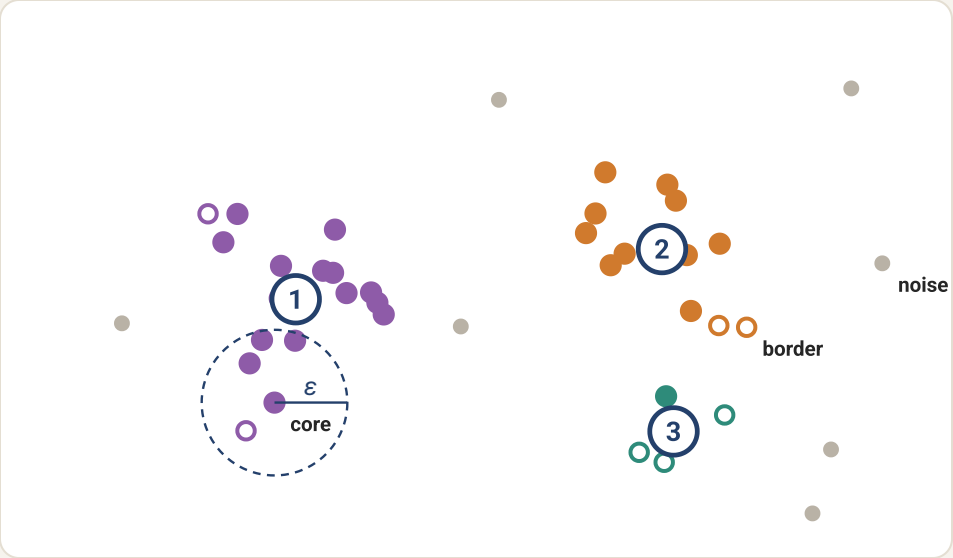
**Not sized**
Delivery levers (third-party; channel margins not disclosed), membership and CRM, waste (no inventory model in this deck), and the new-store site forecast (shown as cups added only).

**ASSUMED BASELINE** illustrative
Flat normal-day roster, all by hand: about **1,291** peak-hour orders a year walk away per outlet.

**PROPOSED SCENARIO** in our model
Both levers, one run: **1,278** kept a year per outlet, profit **S$1.09m** across 44 outlets (S$0.37m without the minimum-crew rule).

# DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

## DBSCAN IN FOUR MOVES · TOY EXAMPLE




1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 05 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

## THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

| | | | |
|---|---|---|---|
| 1 | **App searches** | Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns | **3,000 dots** |
| 2 | **No store nearby** | The nearest outlet is further than the walking distance | **1 km · 46% of dots** |
| 3 | **DBSCAN** | Clusters the no-store-nearby dots | **ε 0.6 km · min 15 · 11 clusters** |
| 4 | **Pull of a store** | Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance | **L = 1 km** |
| 5 | **Who buys where** | A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one) | **not buying 0.3** |
| 6 | **Calibration** | Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days) | **556 cups a day** |
| 7 | **Score a pin** | Cups for Chagee = what the new store sells − what it moves from nearby outlets | **top: 837 − 106 = +731** |
| 8 | **Verdict** | Share of the new store's cups that are new to Chagee | **strong ≥ 60% · weak < 40%** |
| 9 | **Opening roster** | Peak-hour cups ÷ cups one person handles, with the safety margin | **one per 83 cups** |

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

# Every rung, every price as published, and where it comes from

*To our knowledge, based on public sources. Physical AI: automation and machines. A gap is "not disclosed", never a guess.*

| RUNG | EXAMPLE | AUTOMATES · STEPS ON MACHINES (OUR MAPPING) | VENDOR LIST PRICE | CLAIMS (NOT MODEL INPUTS) | RELIABILITY | SOURCE |
|---|---|---|---|---|---|---|
| 1 **Semi-automation** | Gong Cha Singapore | Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled. | *Not disclosed* | Over 20% productivity from self-ordering; 5% manpower saving (no baseline given) | PRESS | https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/ |
| 2 **Beverage platform** | Gong cha Super Wu (Feb 2026) | Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk** | *Not disclosed* | Up to 65% higher peak productivity | COMPANY CLAIM | https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/ |
| 3 **Single automated milk-tea machine** | Amacha (one small shop, owner interview) | Makes the milk tea itself. **Tea and milk** | S$10,000 to buy | The shop's total opening cost was S$209,000 (not a machine cost) | PRESS | https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715 |
| – | **Our scenario: today's split** | **Order and tag, Tea and milk, Shake and seal**. A scenario of ours (slides 06 and 07), not Chagee's disclosed setup | Model: S$1,909 a month per station unit | | OUR ASSUMPTION | assumptions.json |
| 4 **Robot arm that makes boba** | Richtech ADAM | Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal** | *Not disclosed*. Rental only; price not published | – | VENDOR PAGE | https://richtechrobotics.com/solutions/adam |
| 5 **Robot barista** | Oli | A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly. | US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease | – | VENDOR PAGE | https://olirobotics.com/best-robot-barista |
| | Artly | | US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software<br>Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed. | – | VENDOR PAGE | https://olirobotics.com/best-robot-barista |
| | Cafe X | | US$250,000 (≈ S$318,125) to buy<br>Oli's estimate of a competitor's price. | – | VENDOR PAGE | https://olirobotics.com/best-robot-barista |
| – | **Chagee Singapore** | Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment. | *Not public* | – | PRESS | https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/ |

**Vendor list prices,** not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

# Where every number comes from

## FACTS QUOTED

**Outlets in Singapore: 44**. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

**Franchise share of FY2025 net revenue: 0.88**. Secondary source; flagged wherever shown. *[L-koalagains]*

**Company-owned share of FY2025 net revenue: 0.12**. Secondary source; flagged wherever shown. *[L-koalagains]*

## SOURCES

**L-sg-directory** Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-koalagains** KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-yfchuhai** YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-sec20f** Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

**L-unsw** UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

**I-chagee-contact** INFORMAL - a Chagee contact (Oct 2026). Three points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work; Singapore outlets run under one company and one data system; all delivery is via third-party apps. Confirm with Chagee before citing.

**I-sg-groundfloor** INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

**I-team-app-first** INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing.

**L-ester1996** Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 05.

**P-honeycombers** PRESS - in-store sightings, indicative. Honeycombers (Feb 2026), https://thehoneycombers.com/singapore/china-bubble-tea-brand-singapore/ : Chagee price sightings in Singapore; not a current full menu.

**P-eatbook** PRESS - in-store sightings, indicative. Eatbook (Aug 2024), https://eatbook.sg/?p=168476 : Chagee price sightings in Singapore; not a current full menu, and older than the deck.

**I-team-expansion** INFORMAL - team research (Oct 2026): Chagee is looking to expand rapidly in Singapore; how many stores it is going for is confidential. Singapore only. Confirm before citing.

## ASSUMPTIONS (ASSUMPTIONS.JSON, ILLUSTRATIVE)

**Cups per outlet per year**: 200000 cups/store/year

**Share of the day's cups in each open hour**: 0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07 share of daily cups per open hour (10:00 to 22:00), sums to 1

**Busy and event day multipliers**: none 1, moderate 1.5, large 2.5 x forecast arrival rate

**Busiest hours the multiplier lands on**: 2 busiest forecast hours that get the surge

**Seconds per cup by hand, per station**: ticket 5, cup ice 10, tea milk 15, topping finish 12, shake seal 10, handover 8 seconds of one average worker per cup

**Seconds per cup by robot, per station**: ticket 2, cup ice 8, tea milk 10, topping finish 15, shake seal 8, handover 6 seconds per cup per machine unit

**Stations the robots do today**: ticket, tea milk, shake seal

**Spread of worker speeds**: 0.2 +/- share around average speed

**Minute-to-minute variation in pace**: 0.25 coefficient of variation, minute to minute

**Share of new staff**: 0.25 share of roster in first months

**New staff speed**: 0.7 x average speed

**Coordination loss, people around robots**: 0.9 multiplier on human-pool throughput in the mixed setup

**Minimum people on the counter**: all human 2, mixed 1, all machine 0 workers per open hour

**Roster safety margin over the forecast**: 1.2 x forecast peak rate

**Labour cost per hour worked (SGD)**: 15.06 SGD per hour worked, fully loaded (statutory costs)

**Robot cost per station unit per month (SGD)**: 1908.75 SGD per station machine unit per month

**Attendant hours, all-robot counter**: 1 workers per open hour

**Walk-away limit for the GM's split table**: 0.02 share of peak-hour orders that walk away

**Synthetic app-intent dots (slide 05)**: 3000 synthetic app-intent dots on the map

**Walking distance to collect**: 1 km a customer will walk to collect

**Distance decay, share of the walking distance**: 1 distance decay length as a share of walk_km

**Pull of not buying**: 0.3 pull of not buying, against a store next door (= 1)

**DBSCAN cluster radius**: 0.6 km

**DBSCAN minimum searches**: 15 dots within eps to form a cluster core

**Map scale**: 0.033 km per map pixel

**Wait at which a customer walks away**: 5 minutes

**Price per cup, in store (slide 09)**: 5.9 SGD per cup, in store

**Margin per cup, share of price (slide 09)**: 0.6 share of price (contribution per avoided order: price minus ingredients and packaging)

**Busy days a year (slide 09, illustrative)**: 104 busy days a year (day type 'moderate')

**Event days a year (slide 09, illustrative)**: 11 event days a year (day type 'large')

**Walking-back speed (Week 5 units)**: 6 lines per hour

## CLAIMS WE DID NOT MAKE

No Chagee sales, staffing or wait-time data is used; the forecast is not a trained model. We do not claim the bucket brigade always beats fixed stations: with spare staff, a well-assigned fixed line is as good or slightly faster. Machine breakdowns, breaks and order mix are not modelled.

We make no claim about: **Singapore ownership structure** (a Chagee contact says outlets run under one company and one data system, and that the ownership form does not matter here; I-chagee-contact); **delivery profit by channel**; **Chagee's own equipment cost** (not public); **gross margin per cup** (our assumption); **Chagee's current rostering** (unknown; slide 09's baseline is ours).